

PRESENT A COMPLETE SOLUTION FOR EACH PROBLEM. (/50 points)

1. ITERATIVE SCHEMES FOR LINEAR SYSTEMS (/12)

- Show that the general form of the iterative scheme $Qx_{\text{new}} = (Q - \omega A)x_{\text{old}} + \omega b$ for solving the linear equation problem $Ax = b$ can be written as a relaxation scheme $x_{\text{new}} = (1 - \omega)x_{\text{old}} + \omega x_{\text{new}}$. What is the function of the relaxation parameter ω ?
- Differentiate the three iterative schemes in terms of algorithm for solving the next iterate x_{new} and convergence: Richardson, Jacobi, Gauss-Seidel.
- Give the algorithm that optimizes the Gauss-Seidel iteration at run time.

2. INTERPOLATION VS CURVE FITTING (/13)

Given the data below

x	-2	1	2	4
y	-3.2	1.5	2.7	5.2

- obtain the natural cubic spline interpolant
- fit $y = ax + bxe^{-0.125x}$ by solving the associated normal equations
- compare the approxiamtions using (i) and (ii) at $x=3$.

3. CONCEPTS AND DERIVATIONS

- Differentiate Naïve Gaussian Elimination, Gaussian Elimination with partial pivoting, and Gaussian Elimination with full pivoting. When is it preferable to use LU factorization over Gaussian Elimination? (/5)
- How is Newton's interpolation an intermediate method between intepolation using monomial basis and Lagrange method? (/5)
- Derive the algorithm for solving linear curve fitting problem $Ap=y$ using orthogonal transformation. (No need to discuss QR factorization of A.) Why is this method more robust than the method of normal equations? (/5)
- Give the recursive form of Newton's interpolation and derive the formula for Newton's divided differences $\nabla^j y_i$ that solves the coefficients of the interpolant. (/5)